Automatic Weather Station

System Guide

The parts of the system — the sensor, the station PC, its database and web server — how they fit together, and how to install it, step by step.

Observator Automatic Weather Station 1.0.4

19 October 2026

This guide explains the parts of the **Observator Automatic Weather Station** system (version 1.0.4) and how to install it: the software on the station PC that receives the GMX551's readings, stores them and serves the portal.

It is for whoever installs and looks after that PC. Day-to-day use of the portal is in the User Manual.

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1. About this guide

The Observator Automatic Weather Station software runs on one Windows PC at the site. That PC receives the GMX551's readings, stores them, and serves the portal that everyone opens in a web browser. No internet connection is needed.

This guide explains the parts of the system and how they fit together, then walks through the installation. Day-to-day use of the portal is in the User Manual.

It describes version **1.0.4**.

2. The components

The system has five parts. Three of them are software on the station PC, each running as its own Windows service.

Component	Where it runs	What it does
GMX551 sensor	On the mast	Measures wind, temperature, humidity, pressure, dew point, solar radiation and rain. Sends one line of readings every second, in Gill ASCII or NMEA 0183
Network converter	Beside the sensor	Turns the sensor's serial line into a network connection to the station PC
API and sensor stream	Station PC (Windows service)	Listens for the sensor on port 4000. Checks every reading, builds the one-minute and ten-minute values, stores them, and answers the portal's requests on port 3200
Database server	Station PC (Windows service)	MongoDB. Keeps every reading, user, setting and audit entry. Readings are never deleted
Web server (the portal)	Station PC (Windows service)	Serves the portal on port 3201 to any browser on the site network

Nothing else is needed: no internet, no cloud service and no separate server. Everyone else opens the portal in a web browser such as Chrome or Edge.

3. How they connect

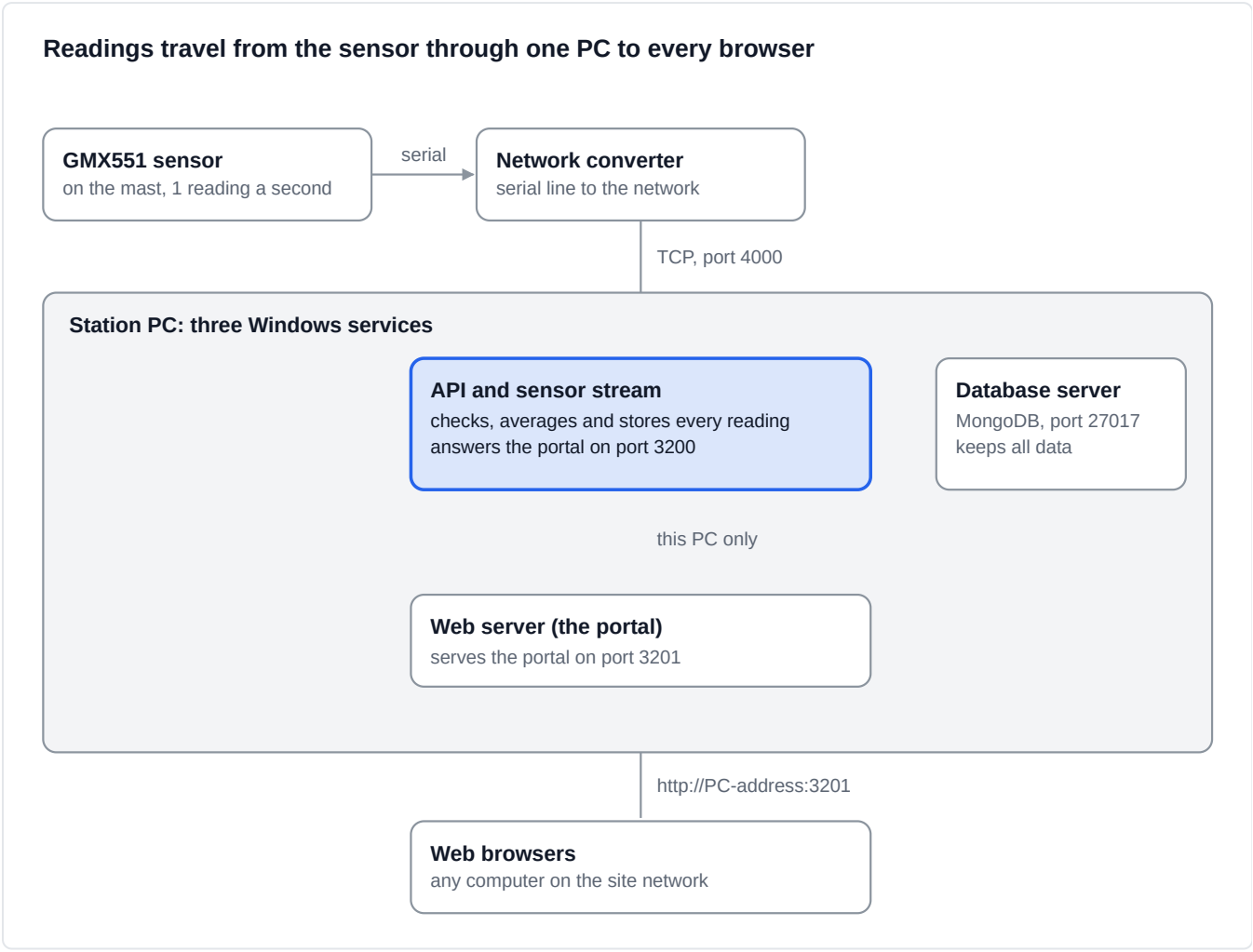


Figure 1 — How the components connect

The sensor's readings enter the PC on port 4000 and are stored by the API. The database (27017) and the API (3200) answer only inside the PC. Only two ports are open to the site network: 4000 for the converter and 3201 for the portal.

4. Where things are on the PC

The program and the data are kept in two separate folders, so an upgrade or an uninstall never touches the readings.

Folder	What is in it
`C:\Observator`	The program, its tools (`status.cmd`, `backup-now.cmd`, `upgrade.cmd` and others) and a copy of the guides. Replaced by an upgrade
`C:\ObservatorData\config\observator.env`	The settings, in one file
`C:\ObservatorData\db`	The database
`C:\ObservatorData\logs`	Logs from each service and each install, rotated automatically
`C:\ObservatorData\backups`	Backups taken with `backup-now.cmd`, and the one an upgrade takes first
`C:\ObservatorData\uploads`	The uploaded logo

The settings and data folders open for administrators only, because the settings file holds the keys that protect sign-in.

The three services run as Windows' built-in *Local Service* account, not as an administrator. Each starts with Windows and restarts by itself after a crash.

5. Before you install

Installation takes about 15 minutes, once, on the one PC that will be the station's server.

You need	Why
Windows 10 or 11, 64-bit	The software is built for it (Windows Server 2019/2022 also work)
An administrator account	Setup installs Windows services and firewall rules
4 GB memory, 20 GB free disk	Readings are never deleted; the data grows by about 0.5 GB a year
A fixed IP address	The converter, and everyone's browser, find the PC by it
Never sleep, time sync on	Every reading is time-stamped by this PC's clock
The setup program	`observator-weather-1.0.4-setup.exe`. No internet is needed

Find the PC's IP address. Press the Windows key, type `cmd` and press Enter. Type `ipconfig` and press Enter. The line **IPv4 Address** is the one, for example `192.168.1.20`.

6. Installing step by step

The setup program asks four questions, then installs and starts everything by itself. For most sites, keep every default.

Step 1 — Start the setup program

Double-click `observator-weather-1.0.4-setup.exe`. Windows asks whether to let it make changes: choose **Yes**. Then select **Next**.

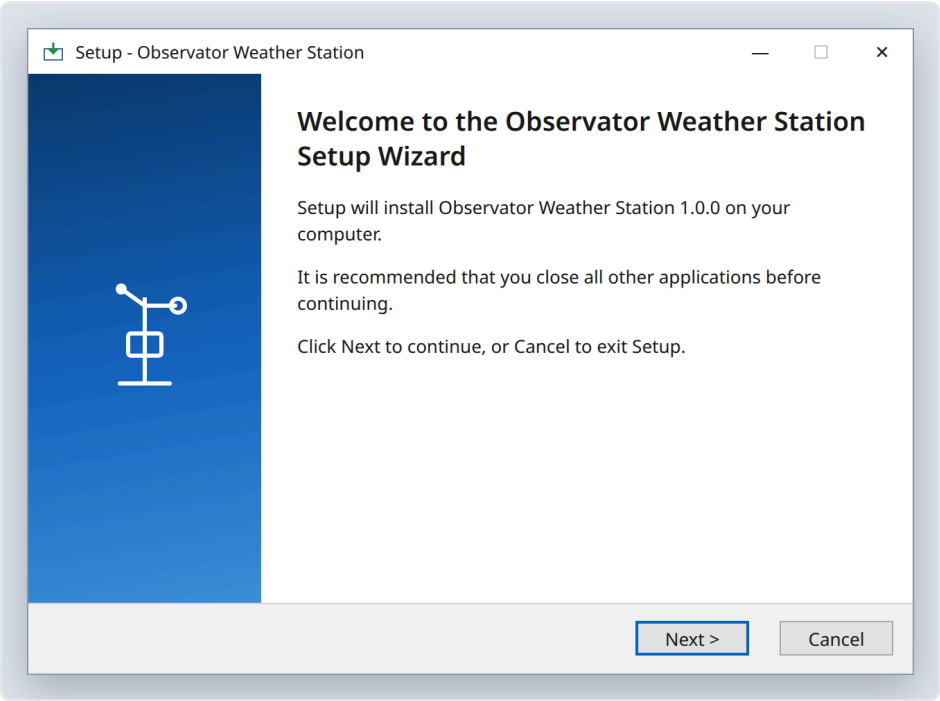


Figure 2 — The setup wizard opens

Step 2 — The program folder

Where the program goes. Keep the default, `C:\Observator`, and select **Next**.

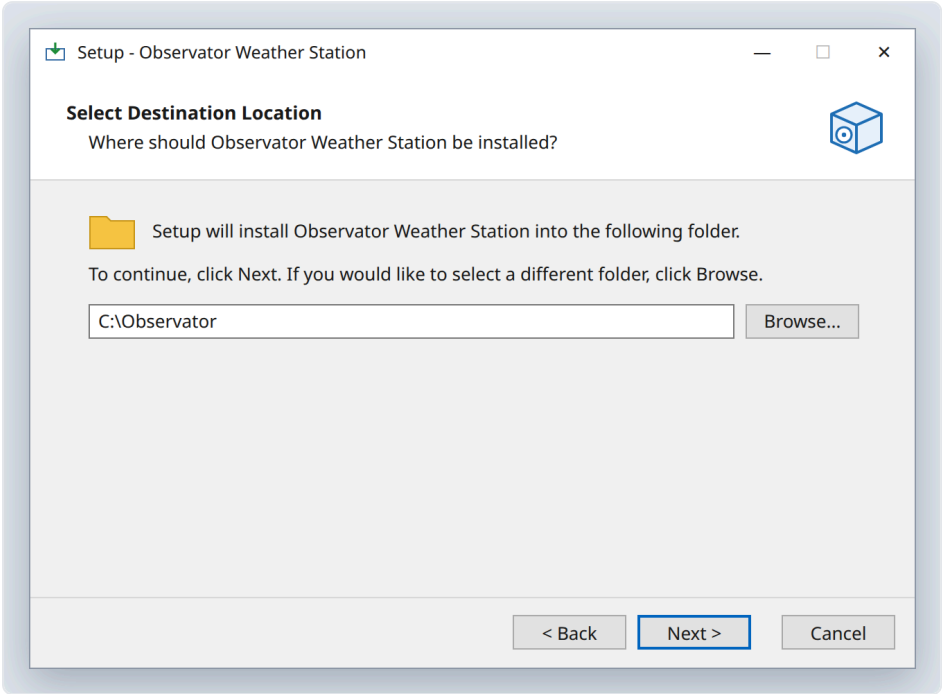


Figure 3 — The program folder

Step 3 — The data folder

Where the readings, settings and logs are kept. Keep the default, `C:\ObservatorData`, and select **Next**. Upgrades and uninstalling never touch this folder. It must not be inside the program folder; setup refuses if it is.

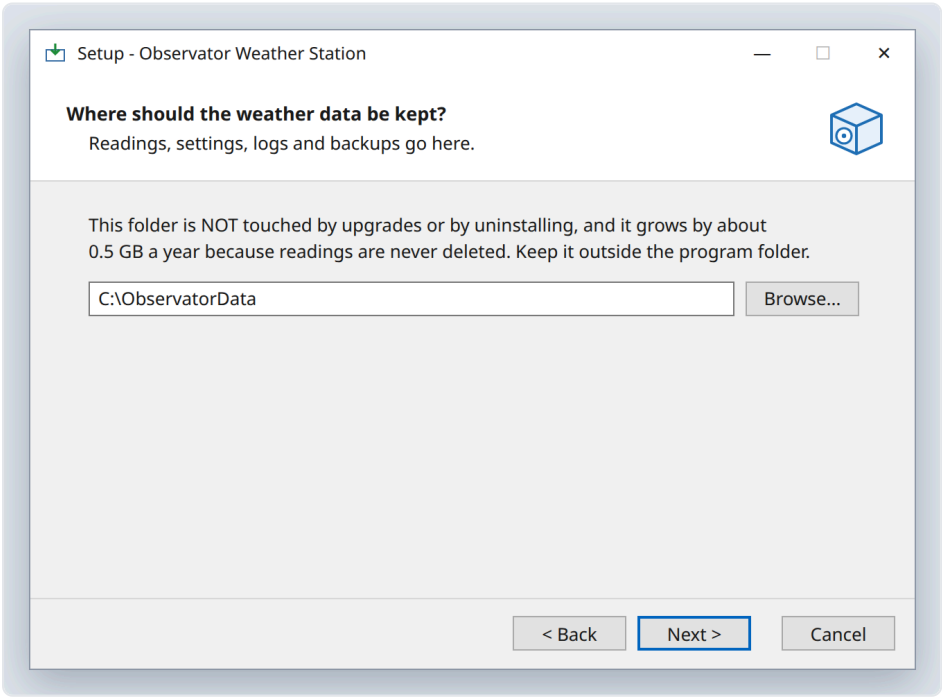


Figure 4 — The data folder

Step 4 — The first administrator

Type the email address of the person who will sign in first and set everyone else up. It does not have to be a real mailbox: the PC sends no email. Select **Next**. The password is asked for later, in its own window, so it is never written down on the PC.

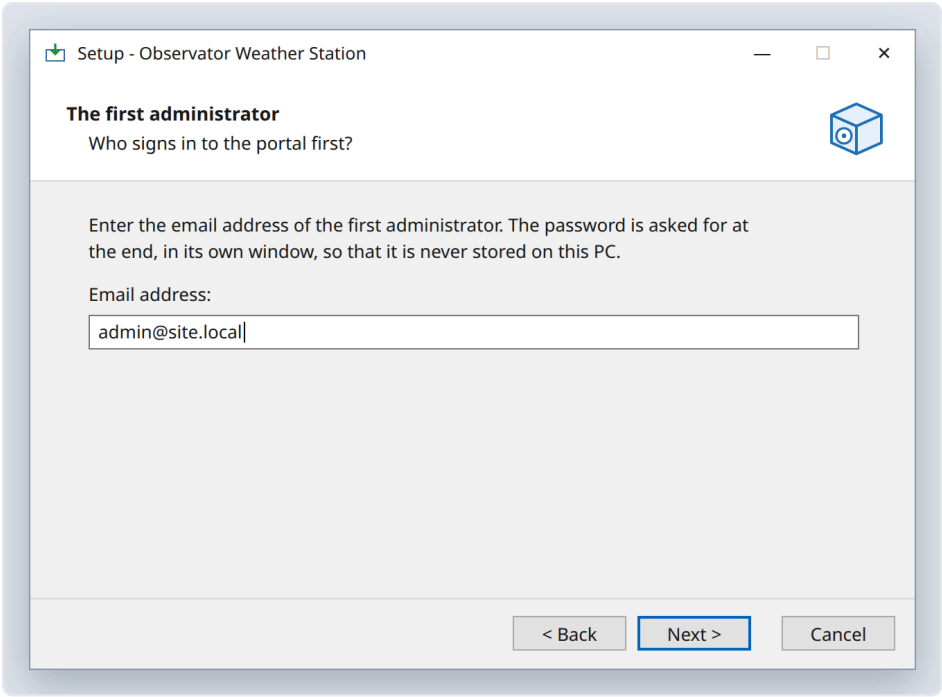


Figure 5 — The first administrator

Step 5 — The sensor and the network

How the GMX551's readings reach this PC. For most sites, change nothing and select **Next**.

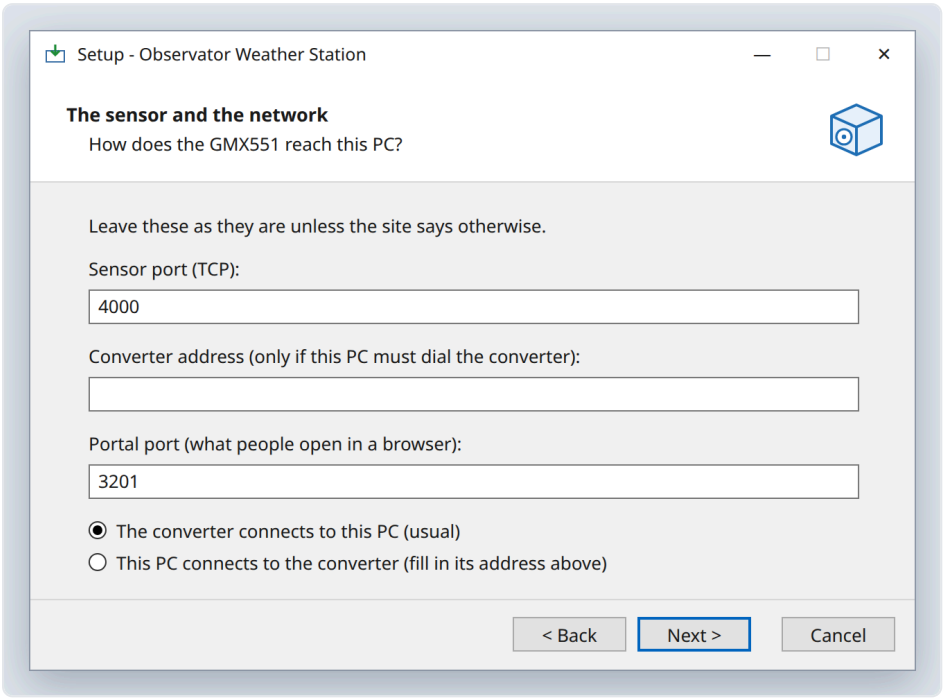


Figure 6 — The sensor and the network

Setting	What to put
Sensor port	`4000`. The converter sends the readings to this port
Converter address	Leave it empty, unless you choose *This PC connects to the converter*
Portal port	`3201`. People open `http://<this PC's IP>:3201`
Converter connects to this PC	The usual choice. Then set the converter to send to this PC's IP, port 4000
This PC connects to converter	Only if the converter can only wait to be called. Then type the converter's IP address

Step 6 — Install

The wizard shows where it will install. Select **Install**. Copying the program takes a minute or two.

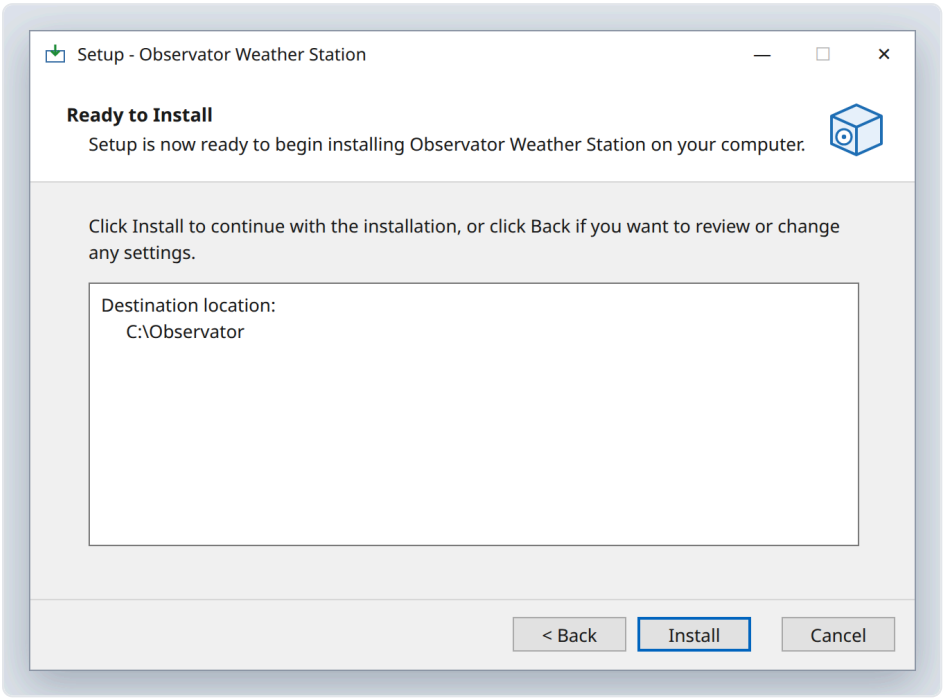


Figure 7 — Ready to install

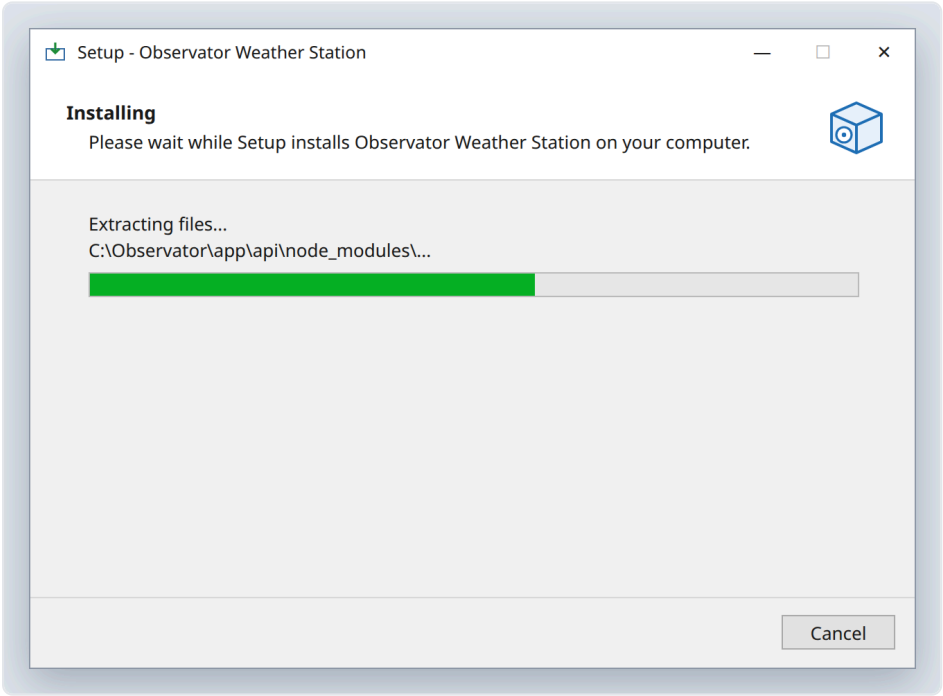


Figure 8 — Copying the program

Step 7 — The administrator's password

A blue PowerShell window opens. It checks the PC, then asks for the first administrator's password, **at least 8 characters, typed twice**.

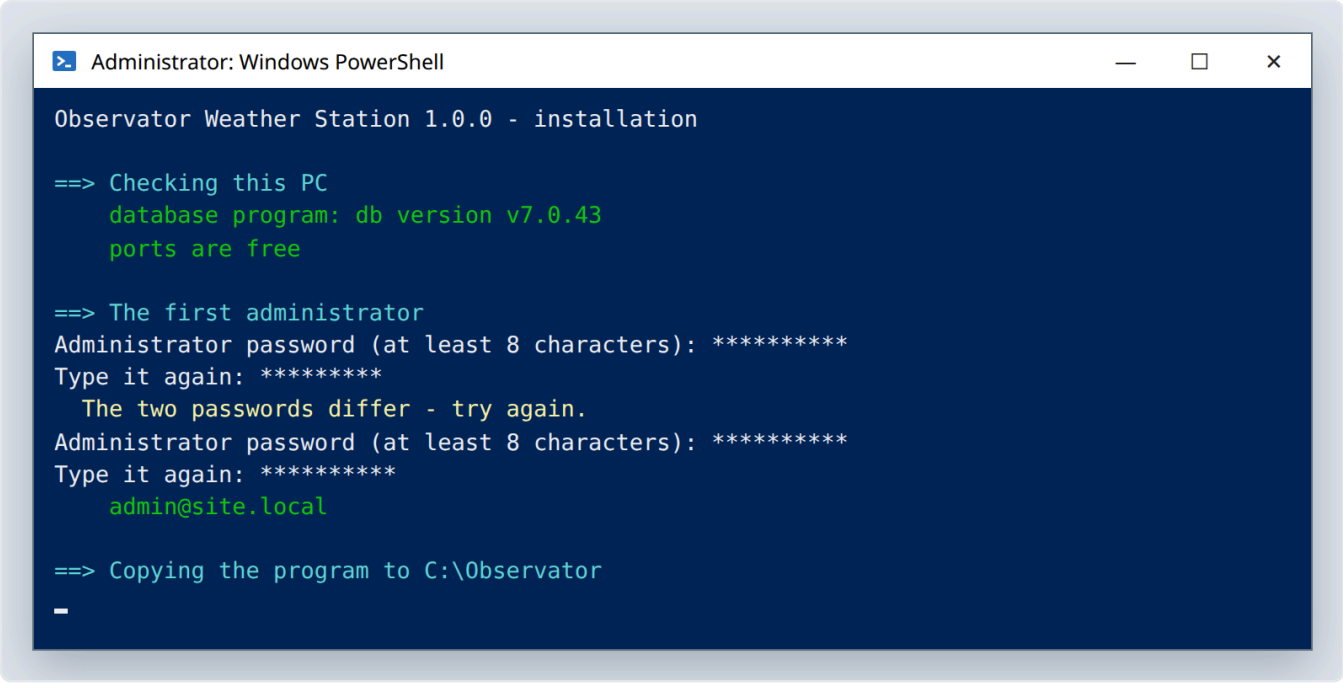


Figure 9 — The password window

- Only stars appear while you type. That is normal.
- If the two do not match, or it is too short, it says so in yellow and asks again.
- Write the password down somewhere safe. You sign in to the portal with this email and password.

Step 8 — Setup finishes the work

The same window carries on by itself. Do not close it while it is working.

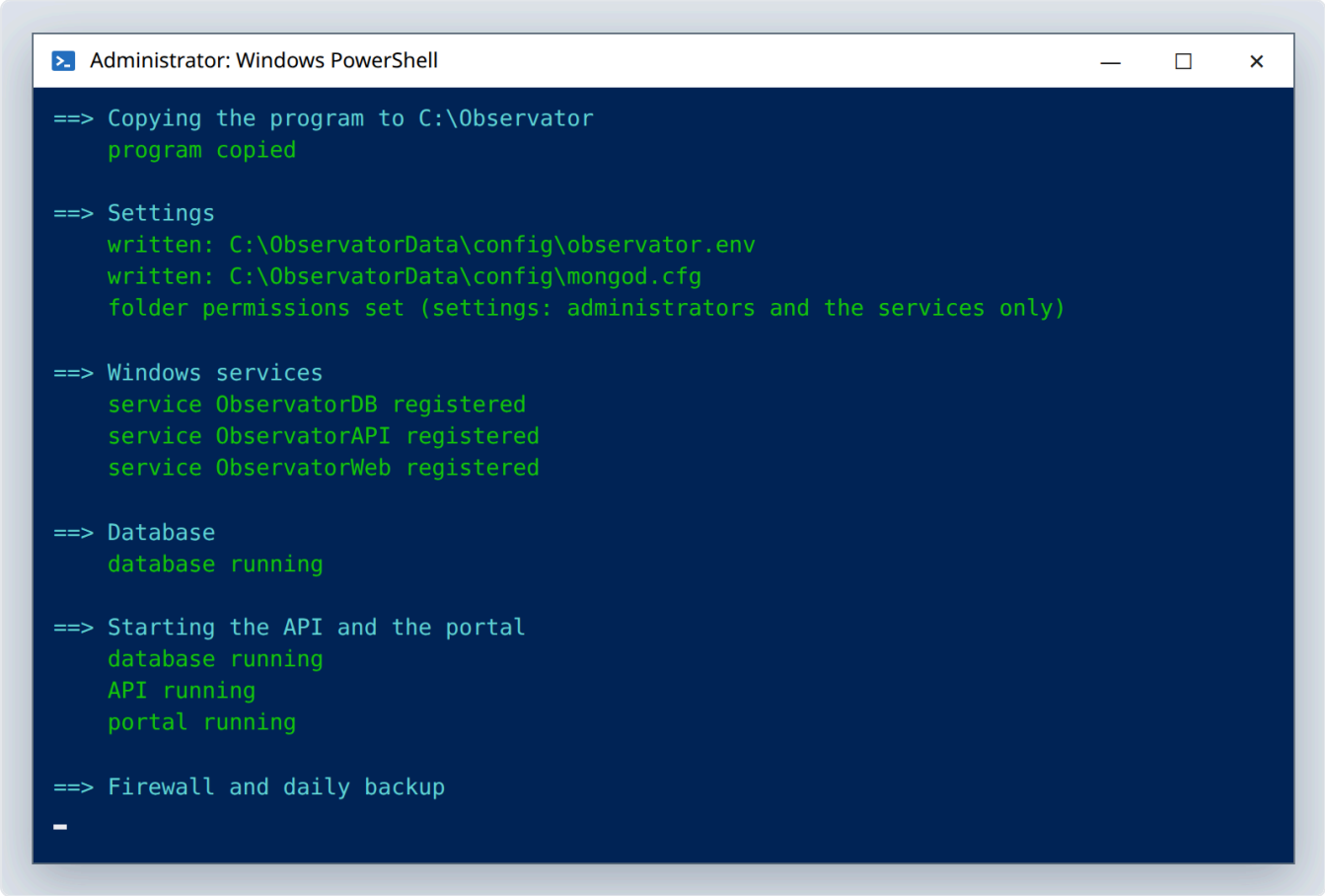


Figure 10 — Setup at work

It writes the settings, creates the three Windows services (database, API and sensor stream, web portal), starts each one and checks it answers, and opens the two ports in the Windows firewall for the local network. When it is finished, the window closes by itself.

Step 9 — Done

Select **Finish**. The software is now in `C:\Observator` and starts by itself whenever the PC starts. The setup program can be deleted.

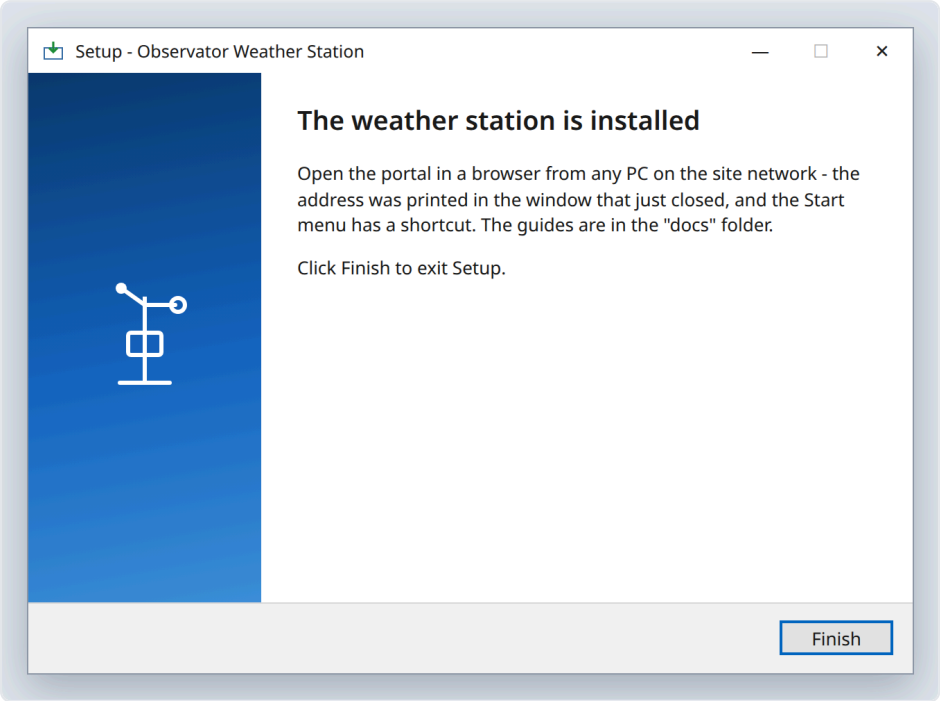


Figure 11 — Setup is finished

7. After installing

Two things remain: open the portal, and point the converter at the PC.

Open the portal

On the station PC, or any computer on the same network, open a web browser and go to `http://<this PC's IP address>:3201`, for example `http://192.168.1.20:3201`. On the station PC itself, `http://localhost:3201` works too, and the Start menu has a **Weather station portal** shortcut.

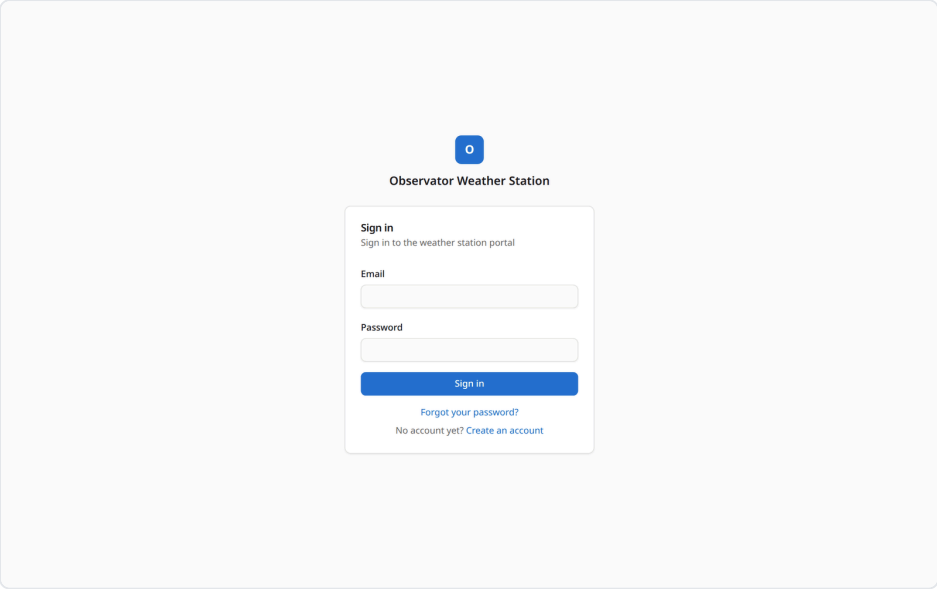


Figure 12 — Signing in

Sign in with the email from step 4 and the password from step 7. Then open **System** in the menu. Everything should be green except the sensor stream, until the converter is set up.

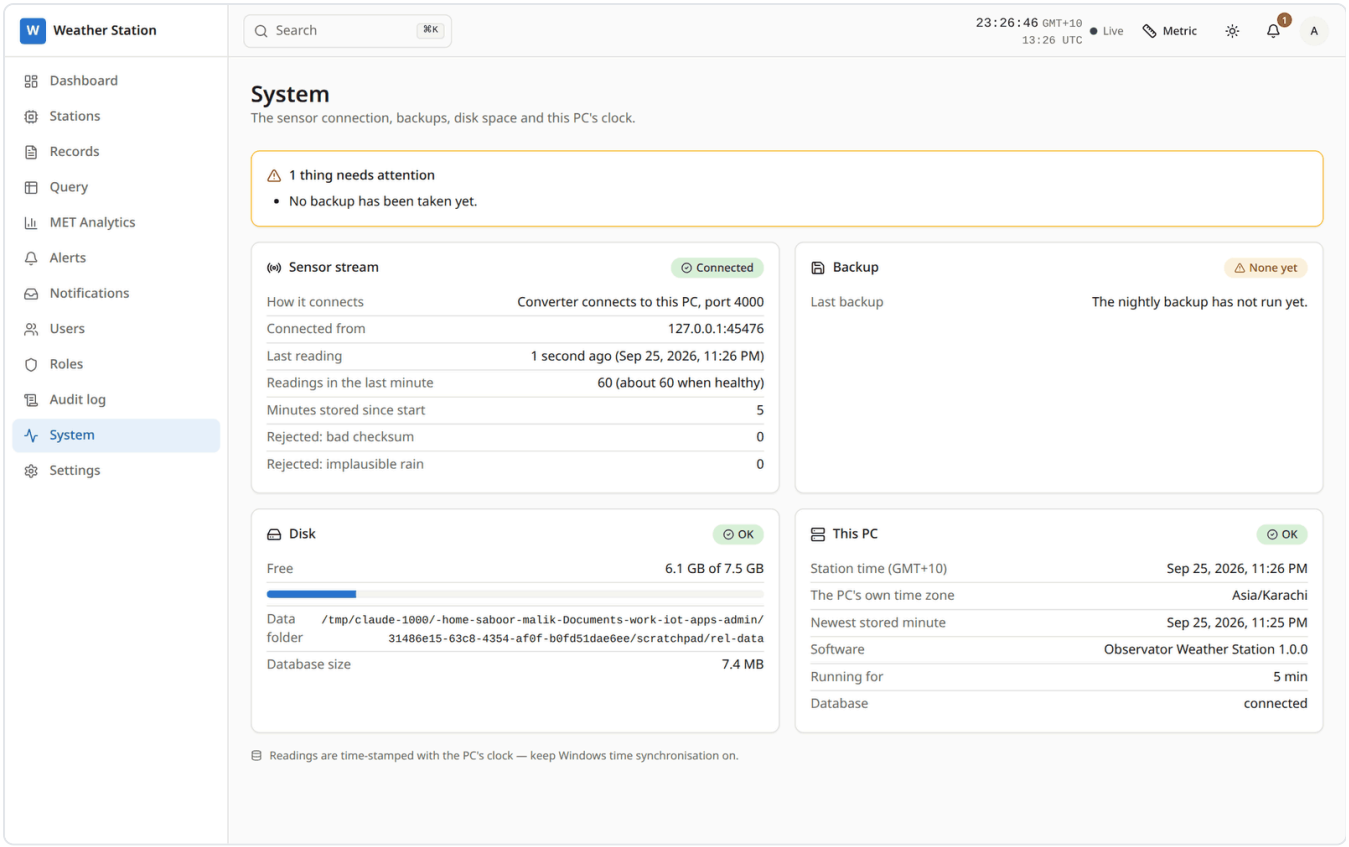


Figure 13 — The System screen

Point the converter at the PC

The GMX551 is wired to a serial-to-network converter. In the converter's own settings page:

Setting	Value
Mode	**TCP Client** (sometimes called *active* or *connect* mode)
Destination IP	This PC's IP address, for example `192.168.1.20`
Destination port	**4000**
Serial	Must match the GMX551. Gill's default is 19200 baud, 8 data bits, no parity, 1 stop bit. A sensor set to NMEA usually uses 4800 or 9600 baud

The software reads Gill ASCII and NMEA 0183, with nothing to set. Within a minute, **System** shows the sensor stream as *Connected* with about 60 readings a minute, and the Dashboard's wind dial starts moving every second.

If the converter can only be a **TCP Server**, the PC calls it instead. Set the converter to TCP Server on port 4000. Then, in the portal, open **System → Sensor stream → Change connection**, choose **This PC connects to the converter**, type the converter's IP address and port, and select **Save and connect**.

8. Running, upgrading and removing

Stopping and starting the weather station

The master administrator (the one created in step 4) sees **Weather station — Stop / Start** on the portal's **System** page.

- **Stop** closes the sensor connection and stops recording. Stored readings are kept, and the PC and portal stay on. It stays stopped after a PC restart, and the System page says so.
- **Start** connects to the sensor again. Recording resumes within a minute.

To restart the services themselves, open **Services** (Win+R, type `services.msc`) and restart the three *Observator Weather* services, or restart the PC.

Tools in the program folder

Tool	What it does
`status.cmd`	Shows whether the services, the sensor and the disk are healthy
`backup-now.cmd`	Takes a backup now, into `C:\ObservatorData\backups`
`restore.cmd`	Puts a backup back
`reset-password.cmd`	Sets a user's password, at this PC
`upgrade.cmd`	In a new release's folder: upgrades, keeping all data
`uninstall.cmd`	Removes the software, keeping the data

Upgrading

Unpack the new release anywhere except `C:\Observator`, then double-click its `upgrade.cmd`. It backs up the database first, keeps the old program as `C:\Observator.previous`, installs the new one and checks it starts. If anything fails, it puts the previous version back and starts it. The data folder is not touched.

Uninstalling

Use **Settings → Apps → Observator Weather Station → Uninstall**. It removes the services and firewall rules and keeps the data in `C:\ObservatorData`. Installing again with the same data folder carries on with all of it.

If setup stops

If something is wrong, the PowerShell window stays open with **SETUP STOPPED** in red and the reason underneath.

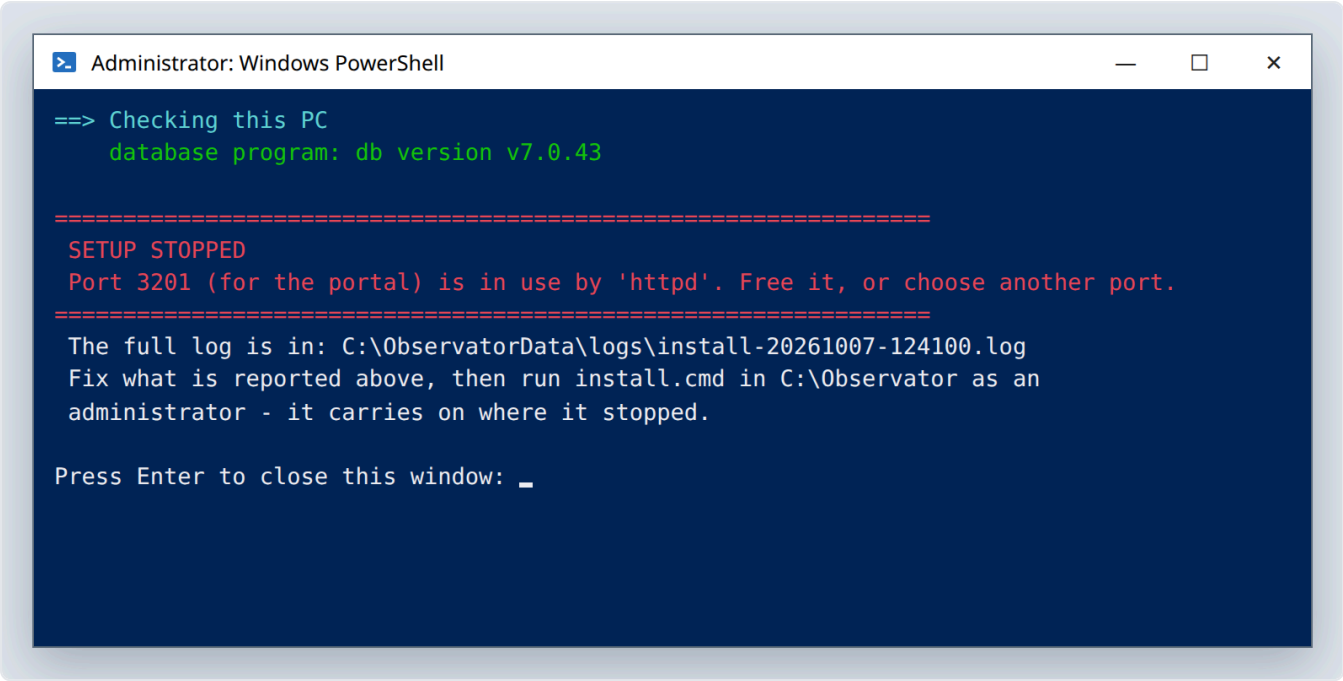


Figure 14 — Setup stopped, with the reason in red

1. Read the red line: it says what is wrong. Take a photo if you need help.
2. Press Enter to close the window. The wizard then shows the message below.
3. Fix what the red line said. Then open `C:\Observator`, right-click `install.cmd` and choose **Run as administrator**. It carries on from where it stopped, keeping the settings you chose.

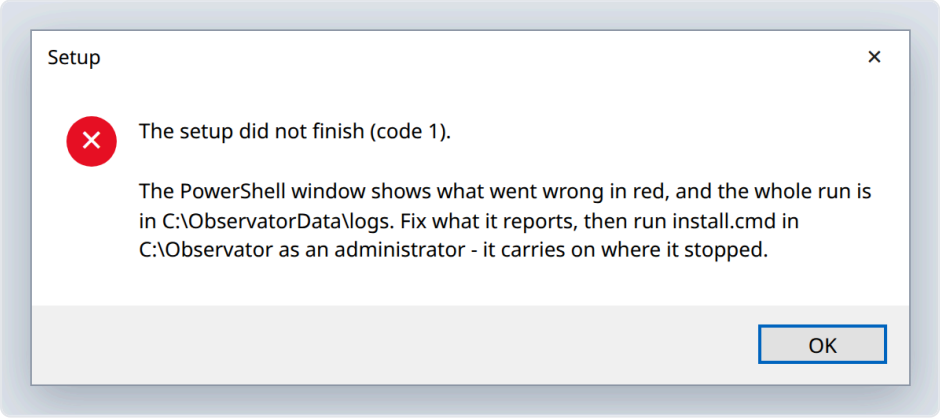


Figure 15 — The message after setup stops

The red line says	What to do
Port 3201 … is in use	Another program uses the portal's port. Stop it, or run setup again with another portal port
Port 4000 … is in use	The same, for the sensor port
… lacks AVX	The PC's processor is too old for the database. Use a newer PC
… did not start	The database, API or portal did not start. Send us the log file it names, from `C:\ObservatorData\logs`
Already installed in …	It is already installed. To update it, use the new release's `upgrade.cmd`

Every run is logged in `C:\ObservatorData\logs\install-<date>-<time>.log`. Send that file if you ask us for help.